Mrinal Mohit

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Technical Skills

Languages: Java, Dart, Python, C++, SQL, JavaScript

Backend & Frameworks: Spring Boot, React, Flask, Flutter

Databases & Cloud: PostgreSQL, MongoDB, Docker, AWS Lambda, AWS ECS, OpenMediaVault

Messaging Systems: RabbitMQ

Developer Tools: Git, GitHub, Maven, Postman

Hardware/IoT: ESP32, GPS Module, Sensors, Arduino

DevOps & Infrastructure: Docker, Jenkins, GitHub Actions, Nginx, Linux, Tailscale

Systems Programming: Linux System Programming, Multithreading, Memory Management

Projects

Self-Hosted Cloud Infrastructure Platform

Tech Stack: *Raspberry Pi 5 — Debian Linux — Docker — OMV — Nextcloud — Tailscale — Nginx — GitHub Actions*

— Engineered a modular self-hosted cloud platform on a **Raspberry Pi 5**, deploying containerized services including **Immich** and **Nextcloud** for private photo backup, file synchronization, and centralized storage management.

— Designed Docker-based infrastructure with persistent volumes, custom bridge networks, restart policies, and reverse proxy routing using **Nginx**, improving service isolation and simplifying multi-service management.

— Configured secure remote connectivity through a **Tailscale** mesh VPN, eliminating public port exposure while enabling encrypted access across multiple devices.

— Built a self-hosted **CI/CD pipeline** using **GitHub Actions** to automate Docker image builds, testing, and deployments, reducing manual deployment effort by approximately **90%**.

— Architected the platform for future scalability with monitoring (**Prometheus/Grafana**), Git hosting (**Gitea**), password management (**Vaultwarden**), and additional containerized services.

GPS-Based Dynamic Toll Collection System *(GitHub)*

Tech Stack: *Spring Boot — Embedded C — PostgreSQL — Redis — RabbitMQ — Docker — Kubernetes*

— Engineered **Embedded C** firmware for **ESP32-WROOM-32** with **NEO-6M GPS** and **SIM800L GSM** modules; implemented **MQTT** client with QoS 1 for reliable telemetry transmission, achieving **3m GPS accuracy** and **99%** message delivery rate

— Built **Spring Boot application** processing real-time GPS data from **ESP32** devices; implemented **PostGIS geofencing** and dynamic pricing with traffic/weather-based multipliers.

— Engineered **JWT authentication**, **Redis**-based rate limiting, and fraud detection achieving **95% accuracy**; deployed event-driven architecture using **RabbitMQ** for async processing.

— Containerized with **Docker/Kubernetes** achieving **98% uptime** and $<$**200ms** response time; integrated **Grafana** monitoring with zero-downtime deployments.

MeetFlow AI – Multi-Agent Enterprise Workflow Automation System *(GitHub)*

Tech Stack: *Python — FastAPI — LangGraph — Gemini 2.0 Flash — PostgreSQL — Redis — Docker*

– Designed a **7-agent LangGraph-based orchestration system** to automate enterprise workflows (Jira, Slack, HR) with **fault-tolerant checkpointing** and audit trails.

– Engineered **production-grade reliability** using circuit breakers, retry logic, rate limiting, and Redis caching, achieving **2–3**$\times$ **throughput improvement**.

– Built **LLM-powered decision extraction and verification pipeline**, enabling autonomous task execution with **human-in-the-loop approval** for high-risk actions.

– Implemented async FastAPI backend with connection pooling and priority queues, supporting **20–30 req/s** and reducing response latency by **30–50%**.

Web & API Automated Validation Framework

Tech Stack: *Java — Burp Suite Extender API — OWASP ZAP — Docker — JWT*

— Developed a **Java**-based security testing framework using **Burp Suite Extender API** and **OWASP ZAP** for automated detection of **OWASP Top 10** vulnerabilities across **Web and REST APIs**.

— Implemented exploit modules for **SQLi, XSS, SSRF, CSRF, host header injection**, parameter pollution, prototype pollution, and insecure deserialization.

— Created PoCs for **JWT tampering** (weak key brute force, `alg=none` abuse), **session fixation**, and **Broken Access Control via IDOR and privilege-escalation fuzzing**.

TEDx Pass Generator – Full-Stack Cloud Application *(GitHub)*
Tech Stack: *Java — Spring Boot — MongoDB Atlas — GCP App Engine — Maven*

— Built enterprise **REST API** with **Spring Boot** using layered architecture, DTO pattern, and centralized exception handling for event pass lifecycle management.

— Engineered automated pass generation with **ZXing (QR codes)** and **iText (PDF)**.

— Implemented secure **MongoDB Atlas**, supporting cloud **deployment on GCP**.

— Deployed on GCP App Engine achieving **99.9% uptime** with **auto-scaling**.

Experience

Software Developer Intern — Mar 2025 — Apr 2025
Know Your Trips, Dublin

— Led a **team of 5 developers**, conducted **15+ code reviews**, and **mentored** team members to deliver **scalable mobile features**, improving sprint velocity by **15%**.

— Integrated **JWT authentication** and **Firebase Cloud Messaging**, strengthening **system reliability** by **30%**.

— **Automated** deployments through **CI/CD**, reducing release cycles by **20%**.

— Contributed to **QA testing** and debugging, enhancing app stability by **25%**.

Tech Stack: Flutter, Dart, GetX, MongoDB, AWS, and Firebase.

Certifications

• AWS Certified Solutions Architect - Associate *(Credential)*

• Oracle Certified Java Professional (Java SE 17) *(Credential)*

• IBM Introduction to Cloud *(Credential)*

Education

Vellore Institute of Technology Bhopal — Bhopal, India
B.Tech in Computer Science and Engineering — *Expected Sept 2027*
CGPA: 8.21/10

Leadership and Impact

Technical Head — Mozilla Firefox Club VIT Bhopal — Feb 2025 — Jul 2025
VIT Bhopal University

— Led 20+ member tech division, mentoring peers in open-source, web, and automation development.

— Conducted coding bootcamps and hackathons impacting **150+ students**.

— Enhanced cross-team collaboration and project execution using agile practices.

Co-Curricular Activities

— Max. rated 1184 on CodeForces.

— Participated in hackathons focusing on fintech automation and cloud-based scalability.